ENGO 531/632

Advanced Photogrammetric and Ranging Techniques

Laboratory Assignment 3

Geometric Calibration of Your (Mobile Phone) Digital Camera

Date given: 9 October 2026

Date due: 22 October 2026 at 23.59
Late group submissions will not be accepted;
Late individual submissions lose 10% per day

Marks: This assignment constitutes 15% of your final grade.

Objective

The objective of this lab is to perform a self-calibration of your own (mobile phone) digital camera and analyze the results. You will design a network within END 305 and collect images and perform the calibration. The calibration results from this lab will be used for lab assignments 4 and 5.

For students in ENGO 531

This assignment is to be performed by the group formed for labs 1 and 2.

For students in ENGO 632

This is an individual assignment.

Assignment

Data for this assignment are not provided. You will collect data with your own digital camera. It is suggested that you use your mobile phone camera since the imagery captured with most phone cameras is largely corrected for lens distortions. If you notice excessive lens distortion in your camera imagery, don't use it—contact the instructor. If you don't have access to a suitable camera, contact the instructor. In lab 4, you will measure an object of your choice with the same camera and use the IOPs you estimate in this lab.

Calibration Network Design

You will use the target field in END 305. Access to END 305 is controlled and will be provided by the TA during the scheduled laboratory period. Object space coordinates for the small targets on the walls of the northwest corner of the room will be provided in D2L. Coordinates are provided for most (but not all) the targets: 213 in total. You do not need to use all of them; 40 to 50 should suffice! The full set is provided to give you flexibility in your network design. Note the format for each row of this file:

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point_id   X   Y   Z
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The units of the coordinates are mm. A plot of the targets in plan view (i.e. X and Y) is often a helpful tool for planning your network. Targets for which coordinates are not available can still be observed and estimated as tie points.

You must design a calibration network and acquire images of the target field. Your network design should consider the zero- and first-order design measures discussed in class. In terms of ZOD, use three weighted control points for datum definition.

We have discussed several FOD measures in the lectures. Environment conditions will likely be such that you cannot exactly achieve the ideal geometric network conditions. The important outcome is that you follow the principles of good network design as best you can within the environment (i.e. space) constraints. If you cannot achieve certain aspects of the design, explain why and what are the impacts on the results. Eight to ten images should be sufficient.

IMPORTANT:
If your camera can capture imagery in different modes, be sure to note the details so that you capture data in the same mode for Lab 4. You should also disable the auto rotate feature of your camera.

Make sure you carefully determine the approximate EOP values of each image. Detailed field notes with a sketch should be produced. High accuracy measurement is not required for position—several centimetres will be fine. The angles only need to be within a few tens of degrees and can be determined by eye, but they must be in the correct quadrant for the adjustment to converge. Both the magnitude and the sign of the angles are often confusing in close-range photogrammetry, so take time to carefully figure out the angles. Pay close attention to the κ rotation angle, specifically whether you rotate the camera clockwise or counterclockwise.

Often the best way to derive the angles is graphically by constructing the rotation matrix **M**. Make a sketch of your coordinate system, the approximate image locations and the axes of image space at each location. See Figure 1 for an example.

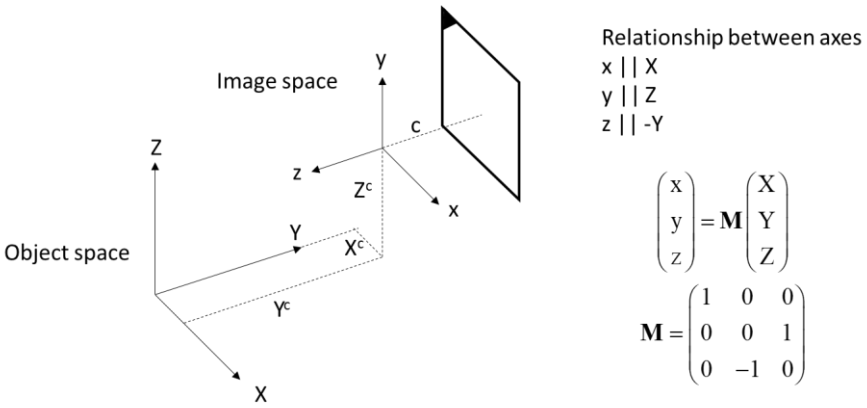


Figure 1. Deriving approximate rotation matrix; (ω^0, ϕ^0, κ^0) = (90°, 0°, 0°)

For hand-held imagery looking at the target field and the coordinate system definition as shown:

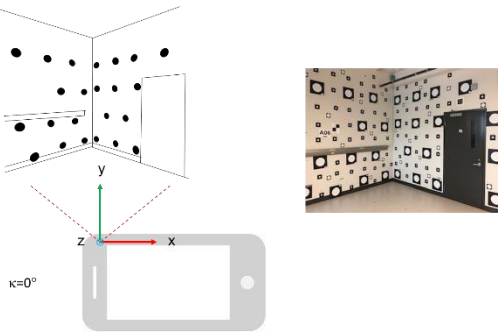
- The ω angle is nominally 90° (it may deviate from this value by $\pm$20°, depending on how much the camera has been inclined or declined);

- The ϕ angle roughly corresponds to the horizontal direction in which the camera's optical axis is pointing (rotation about the y axis in Figure 2; 0°); and
- The κ angle is 0°, -90° or 90° depending on whether the camera is in landscape or portrait mode and in which direction it has been rotated about the z axis.

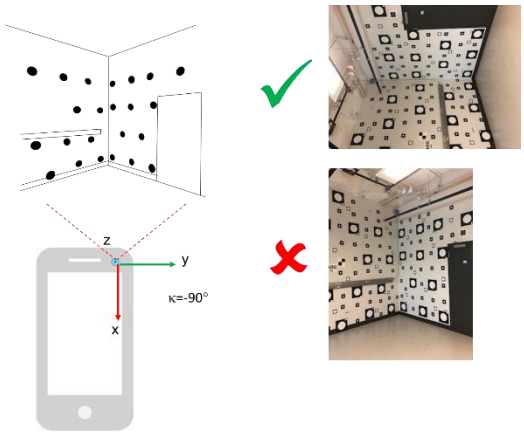
You are strongly advised to review the rotation conventions used in this course prior to image capture.

Measure the coordinates of the target centres in each image. There is no need to develop an algorithm to precisely measure the targets. Manual measurement is fine. Use of a free image viewer such as IrfanView on PC or Gimp on MacOS that allows measurement of image coordinates. To do so in IrfanView, place the cursor over the target, hold down the left mouse button and read the coordinates. Note that these are in a left-handed coordinate system relative to the upper-left corner of the image. Before you proceed to the self-calibration adjustment, it is strongly recommended that you overlay your measured image points with IDs on each image to ensure that the target labelling is correct. This step will take a few extra minutes, but it could save you a lot of time later if you can identify any gross labelling errors before the bundle adjustment.

It is extremely important to maintain orientation of each image as captured. The operating system and/or software may rotate the image for display purposes. The figures below illustrate the problem and how it should be handled.



Landscape orientation maintained as required



Portrait orientation image without rotation, as required (top), and with unwanted orientation (bottom)

Determine the approximate IOPs. The centre of the image format can be used for the principal point. You can derive the principal distance as follows. Mark out two points on the floor and measure the distance between them with a tape measure. Capture an image of these two points holding the camera level (i.e. nadir looking). The two points should fill the image format. Measure the orthogonal distance from the camera to the floor with a tap measure. Measure the distance between the points in the image and compute the principal distance.

It is worth restating that the selection of units is very important in a bundle adjustment. Object space coordinates (control points and PCs) are in mm. Image space units can be in pixels, millimetres or micrometres. Any of these three choices is perfectly fine, but pixels should be used here. In terms of

significant figures, image space quantities (observations, IOPs, etc.) are usually reported to the nearest hundredth of a pixel. Object space coordinates are usually reported to the nearest tenth of a mm. Angles can be reported to a thousandth of a degree.

Perform the self-calibration adjustment. It is suggested that you do this in stages. First, run an initial adjustment with all object points fixed (treating all as GCPs) and the IOPs fixed to the nominal values. If you have done a good job determining the approximate IOPs and EOPs, this adjustment should converge. Note that the residuals may be quite large due to the over-constrained datum and the use of nominal IOPs. If this adjustment does not converge, it is likely due to outliers and/or poor approximate values. In this case, you should break down the adjustment to a smaller subset of images, even a single image, to track down the problem. Once that solution converges, add a second image and so on. When the adjustment with all images converges, run the (nearly) minimally constrained self-calibration adjustment and analyze the results.

You must choose the standard deviation of the image point and control point observations. The former can be judged from how precisely you believe you can observe your targets in the imagery. You can start with a value of 1 pixel (or equivalent in mm or μm) and refine it. However, you should justify your final choice based upon certain bundle adjustment outputs. You should also perform outlier detection and removal.

Submissions

You are to report on the findings of your camera calibration by addressing the following aspects:

- Details of the camera you calibrated
- Geometric network design including ZOD, FOD and SOD measures
- Self-calibration results and analyses
    - Overall summary of the network
    - Network quality (precision and reliability)
    - The estimated IOPs and their quality
    - The completeness of the IOP model

Note that the quantities you analyze are for you to decide, but your analyses are expected to be detailed and justified. It is strongly suggested that you include plots and tables to support your observations. Submissions should include field notes and a sketch of the data capture

Submissions are to be uploaded as a single PDF file to the D2L Dropbox by the specified date and time. For ENGO 531 students, only one group member needs to make a submission but be sure to indicate all members of the group on the title page.

END OF LAB 3